

LETTER TO THE EDITOR

Massive stars in the SDSS-V survey: New O2 stars in the Large Magellanic Cloud

A. Roman-Lopes¹, J. G. Fernández-Trincado⁴, Marina Kounkel⁵, A. A. C. Sander⁶, Carlos G. Román-Zúñiga²,
A. Z. Lugo-Aranda³, Sergio Sánchez-Sanjuán², Abhinna Sundar Samantaray⁶, José Eduardo Méndez Delgado⁷,
and Christian Moni Bidin⁸

- ¹ Departamento de Astronomía, Universidad de La Serena, Av. Raúl Bitrán 1305, La Serena 1700000, Coquimbo, Chile e-mail: aroman@userena.cl
² Instituto de Astronomía, Universidad Nacional Autónoma de México, A.P. 106, Ensenada 22800, Baja California, Mexico
³ Universidad Andrés Bello, Facultad de Ciencias Exactas, Departamento de Física y Astronomía – Instituto de Astrofísica, Autopista Concepción–Talcahuano 7100, Talcahuano, Chile
⁴ Centro de Investigación en Astronomía, Facultad de Ingeniería, Ciencia y Tecnología, Universidad Bernardo O’Higgins, Av. Viel 1497, Santiago 8370993, Chile
⁵ Department of Physics and Astronomy, University of North Florida, 1 UNF Dr, Jacksonville, FL 32224, USA
⁶ Zentrum für Astronomie der Universität Heidelberg, Astronomisches Rechen-Institut, Mönchhofstr. 12–14, 69120 Heidelberg, Germany
⁷ Instituto de Astronomía, Universidad Nacional Autónoma de México, A.P. 70-264, 04510 Mexico City, Mexico
⁸ Instituto de Astronomía, Universidad Católica del Norte, Av. Angamos 0610, Antofagasta, Chile e-mail: cmoni@ucn.cl

Received ...; accepted ...

ABSTRACT

Context. O2 stars define the hottest end of the normal O-star sequence, but their identification requires blue-optical nitrogen diagnostics absent from many surveys.

Aims. We reassess three luminous Large Magellanic Cloud (LMC) sources with multi-epoch Sloan Digital Sky Survey V (SDSS-V) spectroscopy obtained with the Baryon Oscillation Spectroscopic Survey (BOSS) spectrographs: [WSI2008] 778, MGSD LH 117-43A/140, and Sk –69 135.

Methods. We measure visit-resolved radial velocities and H, He, and N equivalent widths (EWs), compare robust combinations with LMC O2 templates at a common effective resolution, and use OSTAR2002 spectral energy distribution (SED) fits and a Gaia colour–magnitude diagram (CMD) as broad-band checks.

Results. The weighted log [EW(N IV λ 4058)/EW(N III λ 4640)] values are 0.64 ± 0.29 , 0.48 ± 0.23 , and 0.34 ± 0.44 for [WSI2008] 778, MGSD LH 117-43A/140, and Sk –69 135. The detection of N V λ 4604,4620 supports the O2 If* and O2 V–III classifications adopted for [WSI2008] 778 and MGSD LH 117-43A/140, respectively. For Sk –69 135, the uncertain N IV/N III ratio and non-detection of N V motivate O2–3 V–III. Weak He I λ 4471 may indicate unresolved later-type contamination in [WSI2008] 778 and Sk –69 135. The radial velocities of Sk –69 135 show suggestive but non-significant variability. OSTAR2002 fits over 45–55 kK and the Gaia CMD place all three sources in the luminous very-early-O regime; [WSI2008] 778 shows a strong mid-infrared (MIR) excess, MGSD LH 117-43A/140 is nearly photospheric, and Sk –69 135 shows only a modest 3–8 μ m excess.

Conclusions. The BOSS spectra add [WSI2008] 778 and MGSD LH 117-43A/140 to the LMC very-early-O population and identify Sk –69 135 as an additional O2–3 source. These cases show how incomplete blue coverage and ambiguous catalogue associations can conceal the hottest massive stars.

Key words. stars: early-type – stars: massive – stars: emission-line – stars: individual: [WSI2008] 778 – stars: individual: MGSD LH 117-43A/140 – stars: individual: Sk –69 135

1. Introduction

O2 stars define the hottest end of the normal O-star sequence and are rare benchmarks for hot-star atmosphere, wind, and evolution models (Walborn et al. 2002; Crowther and Walborn 2011; Martins et al. 2002; Kudritzki 2002). Their blue-optical classification is set chiefly by the N IV λ 4058/N III λ 4634–4642 balance: N IV strongly dominates N III, He I λ 4471 is absent or very weak, and detected N V λ 4604,4620 further supports the extreme ionisation; He II λ 4686 and the Balmer lines primarily constrain wind morphology and luminosity class (Walborn et al. 2002; Rivero González et al. 2012; Crowther and Walborn 2011). Missing blue coverage, blending,

and nebular emission can therefore produce substantially later classifications.

We reassess three Large Magellanic Cloud (LMC) sources with multi-epoch Sloan Digital Sky Survey V (SDSS-V) spectroscopy obtained with the Baryon Oscillation Spectroscopic Survey (BOSS) spectrographs. [WSI2008] 778 was previously linked to young stellar object (YSO) and post-asymptotic giant branch (post-AGB) selections (Whitney et al. 2008; van Aarle et al. 2011); MGSD LH 117-43A/140 was classified as O3 III(f*) and later as O5 V based on Multi Unit Spectroscopic Explorer (MUSE) data lacking the decisive blue nitrogen lines (Garman and Walborn 1987; Massey et al. 1989;

McLeod et al. 2019); and Sk −69 135 was catalogued only as an OB source (Sanduleak 1970). Our analysis combines visit-resolved spectroscopy, LMC O2 template comparisons, and broad-band checks.

2. Observations and analysis

The southern SDSS-V Milky Way Mapper (MWM) spectra obtained with BOSS cover the blue classification region and extend beyond H α (Bowen and Vaughan 1973; Smee et al. 2013; Kollmeier et al. 2026). Four visits are available for [WSI2008] 778 and three each for MGSD LH 117-43A/140 and Sk −69 135. Each visit was rest-frame corrected with photospheric He II, locally normalised, and measured independently; equivalent widths (EWs) were combined using inverse-variance weights, with positive values denoting absorption. Robust visit combinations were used only for display, while quantitative measurements remained visit-resolved (Appendix A).

The effective resolution was measured directly from the BOSS WRESL arrays rather than from the nominal instrumental range. Across the three sources, R spans about 990–1374 at 4000 Å and 1322–1757 at 5060 Å. For Fig. 1, the science spectra and BI 237 (O2 V), 30 Dor 016 (O2 III–If*), and Brey 77 (O2 If*) were matched to the same wavelength-dependent effective resolution; the EW and velocity measurements use the original spectra. Source-specific values and the convolution procedure are given in Appendix B.

3. Spectral classifications

All three targets show the high-ionisation morphology of the earliest O stars: N IV $\lambda 4058$ is in emission, N III $\lambda \lambda 4634$ –4642 is weak, and He II absorption is prominent (Fig. 1). The weighted $\log |\text{EW}(\text{N IV } \lambda 4058)/\text{EW}(\text{N III } \lambda 4640)|$ values are 0.64 ± 0.29 , 0.48 ± 0.23 , and 0.34 ± 0.44 for [WSI2008] 778, MGSD LH 117-43A/140, and Sk −69 135; N V $\lambda \lambda 4604$, 4620 is detected in the first two but not significantly in Sk −69 135. Full EWs and statistical caveats are given in Table C.1 and Appendix C. Narrow nebular [O III] in [WSI2008] 778 and MGSD LH 117-43A/140 indicates partial H β infilling and possible H γ core contamination, so these Balmer cores are not used as purely stellar diagnostics; He II $\lambda 4686$ provides the cleaner wind comparison.

For [WSI2008] 778, the strong N IV/N III imbalance and detected N V establish O2. Its strong He II $\lambda 4686$ emission places it in the If* morphology, so we adopt O2 If*. Reproducible He I $\lambda 4471$ (0.31 ± 0.10 Å) is stronger than expected for an isolated O2 spectrum and may indicate an unresolved later-type contribution, which cannot be decomposed at BOSS resolution.

For MGSD LH 117-43A/140, N IV strongly dominates weak N III and both N V components are detected, securely supporting O2. He II $\lambda 4686$ remains in absorption; however, the modest BOSS resolving power ($R \approx 1170$ –1260 across the main blue diagnostics) and nebular H β infilling prevent a robust V/III distinction. We therefore adopt O2 V–III. This also explains the previous O5 V classification based on MUSE data, whose wavelength coverage begins near 4750 Å and excludes the defining blue N sequence.

For Sk −69 135, N IV emission and weak N III establish the very-early-O regime and the absorption-dominated spectrum remains close to BI 237. The uncertain N IV/N III ratio and non-detection of N V nevertheless motivate the more conservative O2–3 V–III classification. Weak He I $\lambda 4471$ (0.22 ± 0.09 Å) is reproducible and may likewise reflect a later-type contribution.

4. Kinematics and physical context

The weighted stellar velocities are 295 ± 17 , 240 ± 7 , and 277 ± 7 km s $^{-1}$ for [WSI2008] 778, MGSD LH 117-43A/140, and Sk −69 135. Local nebular [O III] gives 280.8 ± 4.3 , 251.3 ± 4.4 , and 281 ± 19 km s $^{-1}$, respectively, so none of the stellar-nebular differences is significant. For Sk −69 135, three visits over 23 days show a first-to-last change of -39.2 ± 17.8 km s $^{-1}$; a constant-velocity fit gives $\chi^2 = 5.33$ for two degrees of freedom ($p = 0.069$), suggestive but insufficient to establish binarity.

The targets are not concentrated around 30 Doradus: their projected separations from R136 are about 0.66, 1.14, and 0.93 kpc for [WSI2008] 778, MGSD LH 117-43A/140, and Sk −69 135. In particular, MGSD LH 117-43A/140 lies in the LH 117/NGC 2122 association within the N180 H II complex (Massey et al. 1989; McLeod et al. 2019). Thus they sample distinct massive-star environments across the LMC, consistent also with the agreement between stellar and local nebular velocities.

As an independent broad-band check, we fit selected optical–near-infrared photometry with half-solar OSTAR2002 atmospheres. Because these bands sample the Rayleigh–Jeans side of such hot spectral energy distributions (SEDs), the photometry does not determine a unique T_{eff} ; instead, the spectroscopically motivated 45–55 kK grid is sampled with equal weight (Appendix D). The resulting positions in the Gaia colour–magnitude diagram (CMD), $(G_{\text{BP}} - G_{\text{RP}})_0$, $M_{G,0}$, are $(-0.509, -4.97)$, $(-0.514, -5.56)$, and $(-0.478, -5.79)$ for [WSI2008] 778, MGSD LH 117-43A/140, and Sk −69 135, respectively, compared with $(-0.524, -5.40)$ for BI 237 and $(-0.562, -6.08)$ for 30 Dor 016. All three targets therefore lie in the same luminous blue part of the CMD as established very-early-O stars. Figure 2 shows PAdova and TRieste Stellar Evolution Code (PARSEC) v1.2S $Z = 0.006$ isochrones from 0.05 to 2 Myr and an $A_V = 0.2$ reddening vector; these curves are shown only as a guide, not to provide precise ages or masses.

Hot stripped helium stars can overlap very-early-O stars in intrinsic colour, but the observed Magellanic-Cloud sample is much fainter optically (Götberg et al. 2018; Götberg et al. 2023). Stars 1–8 span $T_{\text{eff}} \approx 57$ –90 kK and $\log L/L_{\odot} \approx 3.5$ –5.1; however, their optical absolute magnitudes lie in the range $-1.7 \lesssim M_{V,\text{AB}} \lesssim +0.8$. The optically brightest, Star 1, is shown in Fig. 2. The science targets lie several magnitudes above this regime, so luminosity rather than intrinsic colour is the main discriminant. Their CMD positions strongly disfavour an ordinary intermediate-mass stripped star as the optically dominant component while not excluding past binary interaction, unresolved companions, or more massive binary products.

The OSTAR2002 scaling independently points to the same luminosity regime. The numerical radius/extinction posteriors are reported in Appendix D; qualitatively, [WSI2008] 778 shows a strong rising mid-infrared (MIR) excess, MGSD LH 117-43A/140 remains nearly photospheric, and Sk −69 135 shows only a modest 3–8 μm excess.

5. Discussion

The spectroscopic and broad-band results constrain different aspects of the problem. The adopted subtypes are morphological, based on the N IV–N III balance, N V where measurable, and the wind-sensitive He II/Balmer morphology; the 45–55 kK SED interval is a physically motivated nuisance range for luminosity scaling, not an independent photometric T_{eff} measurement. This distinction matters at the earliest O subtypes, where the mapping

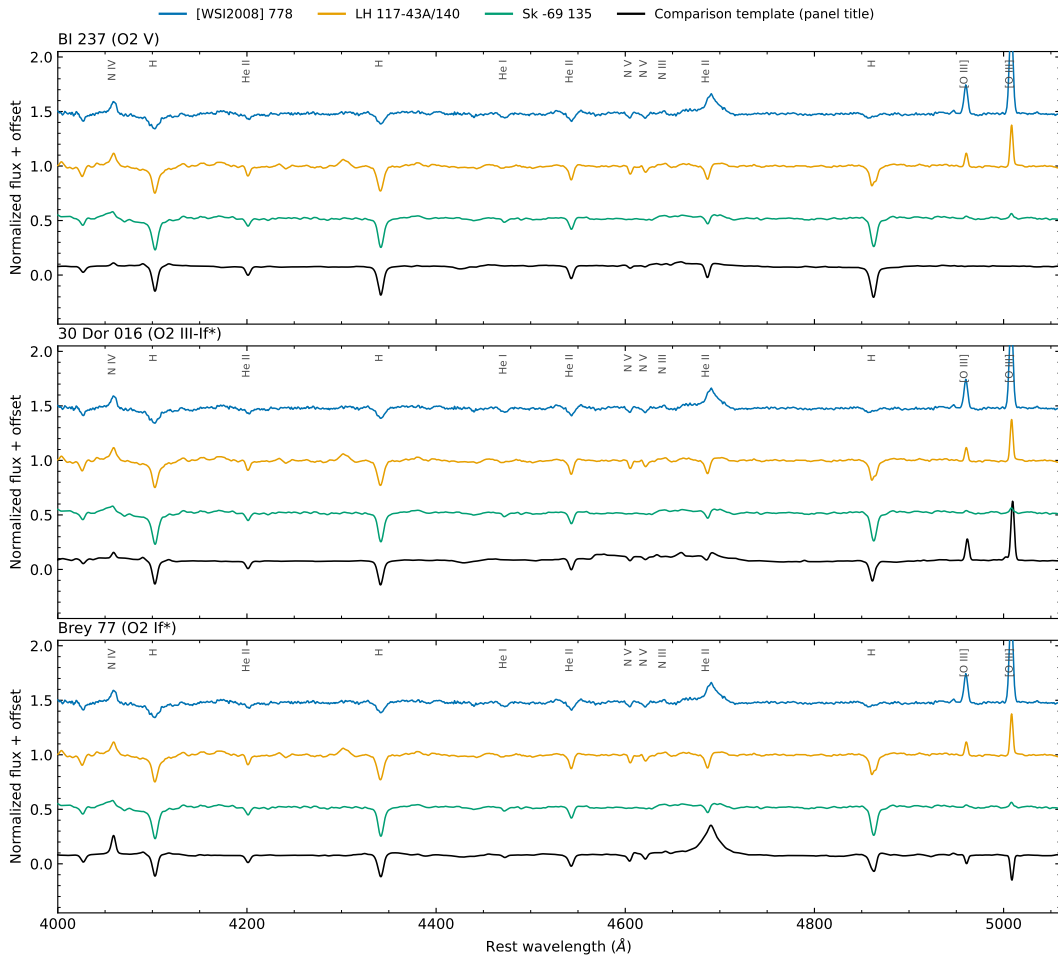


Fig. 1. Rest-frame normalised spectra of the three targets and LMC O2 templates. For this morphological comparison only, all spectra were matched to the same wavelength-dependent effective resolution defined by the largest BOSS WRESL among the science spectra. Narrow nebular [O III] $\lambda\lambda 4959, 5007$ emission in [WSI2008] 778 and MGSD LH 117-43A/140 is accompanied by partial Balmer-line infilling.

between morphology and T_{eff} also depends on gravity, wind density, metallicity, and abundance pattern.

Although all three sources show similarly high ionisation, their wind morphologies differ. [WSI2008] 778 combines O2 nitrogen diagnostics with strong He II $\lambda 4686$ emission, whereas MGSD LH 117-43A/140 retains absorption at this line and cannot be securely assigned to luminosity class V or III at BOSS resolution. For Sk -69 135, N IV emission and weak N III establish the very-early-O regime, but the uncertain ratio and absence of significant NV motivate O2-3 rather than an unqualified O2 subtype.

Even if some of the targets are multiple, a luminous very-early-O component is still required. Reproducible He I in [WSI2008] 778 and Sk -69 135 may reflect cooler companions or fibre contamination, and the Sk -69 135 radial-velocity trend is suggestive but not significant. Such effects can modify line depths and inferred radius scales, but do not naturally reproduce the high-ionisation morphology and optical luminosity with an ordinary intermediate-mass stripped star as the dominant component. Higher spectral resolution, higher signal-to-noise ratio (S/N), and denser time sampling are required to establish multiplicity and quantitative atmosphere parameters.

6. Conclusions

SDSS-V/BOSS spectroscopy establishes [WSI2008] 778 and MGSD LH 117-43A/140 as new members of the LMC very-early-O population and identifies Sk -69 135 as an additional O2-3 V-III source. We classify the three objects as O2 If*, O2 V-III, and O2-3 V-III, respectively. Independently, the Gaia CMD and OSTAR2002 scaling place them in the luminous massive-star regime, several magnitudes above the observed intermediate-mass stripped-star locus, consistent with their identification as massive stars.

The earlier classifications also show why such rare stars can remain unrecognised: [WSI2008] 778 was obscured by its infrared-selected associations, the MUSE classification of MGSD LH 117-43A/140 lacked the decisive blue nitrogen region, and no detailed blue-optical classification was available for Sk -69 135. Complete coverage of N IV $\lambda 4058$, NV $\lambda\lambda 4604, 4620$, and N III $\lambda\lambda 4634-4642$ is therefore essential for recovering the hottest normal O stars. SDSS-V/MWM is well suited to a systematic Magellanic-Cloud search; higher-resolution follow-up will refine wind morphology and multiplicity.

References

I. S. Bowen and Jr. Vaughan, A. H. *Applied Optics*, 12(7):1430–1435, 1973. .

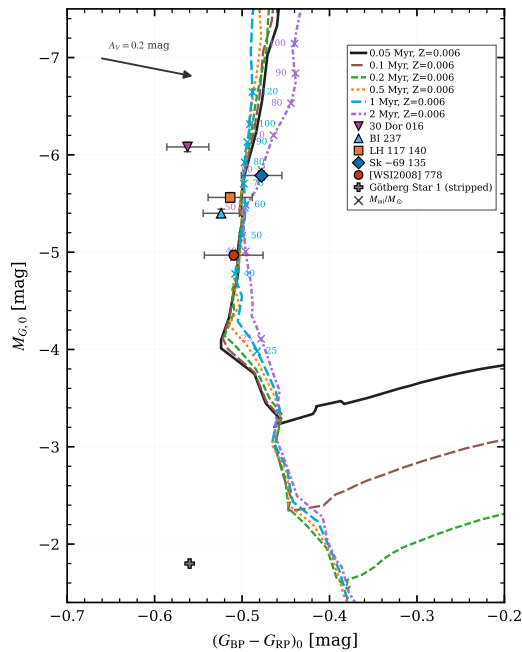


Fig. 2. Dereddened Gaia CMD for the three targets and BI 237/30 Dor 016. PARSEC v1.2S $Z = 0.006$ isochrones span 0.05–2 Myr; the arrow gives the displacement for $A_V = 0.2$ mag. Götberg Star 1 is the optically brightest stripped-star comparison; O-star error bars show the 16th–84th percentile Monte-Carlo intervals.

- A. Bressan, P. Marigo, L. Girardi, B. Salasnich, C. Dal Cero, S. Rubele, and A. Nanni. *MNRAS*, 427:127–145, 2012. .
- P. A. Crowther and N. R. Walborn. *MNRAS*, 416:1311–1323, 2011. .
- Gaia Collaboration. *A&A*, 674:A1, 2023.
- C. D. Garmany and N. R. Walborn. *Publications of the Astronomical Society of the Pacific*, 99:240–244, 1987.
- K. D. Gordon, G. C. Clayton, K. A. Misselt, A. U. Landolt, and M. J. Wolff. *ApJ*, 594:279–293, 2003. .
- Y. Götberg, S. E. de Mink, J. H. Groh, T. Kupfer, P. A. Crowther, E. Zapartas, and M. Renzo. *A&A*, 615:A78, 2018. .
- Y. Götberg, M. R. Drout, A. P. Ji, J. H. Groh, B. A. Ludwig, P. A. Crowther, N. Smith, A. de Koter, and S. E. de Mink. *ApJ*, 959(2):125, 2023. .
- D. Kato, Y. Ita, T. Onaka, T. Tanabe, T. Shimonishi, I. Sakon, H. Kaneda, A. Kawamura, T. Wada, F. Usui, B.-C. Koo, M. Matsuura, and H. Takahashi. *AJ*, 144:179, 2012. .
- J. A. Kollmeier, H.-W. Rix, C. Aerts, James Aird, Pablo Vera Alfaro, Andrés Almeida, Scott F. Anderson, et al. *The Astronomical Journal*, 171(1):52, 2026. .
- R. P. Kudritzki. *ApJ*, 577:389–408, 2002.
- T. Lanz and I. Hubeny. *ApJS*, 146:417–441, 2003. .
- F. Martins, D. Schaerer, and D. J. Hillier. *A&A*, 382:999–1004, 2002. .
- P. Massey. *ApJS*, 141:81–122, 2002. .
- P. Massey, C. D. Garmany, M. Silkey, and Kathleen DeGioia-Eastwood. *The Astronomical Journal*, 97:107–130, 1989. .
- A. F. McLeod, J. E. Dale, C. J. Evans, A. Ginsburg, J. M. D. Kruijssen, E. W. Pellegrini, S. K. Ramsay, and L. Testi. *Monthly Notices of the Royal Astronomical Society*, 486(4):5263–5288, 2019. .
- M. Meixner, K. D. Gordon, R. Indebetouw, et al. *AJ*, 132:2268–2288, 2006. .
- J. G. Rivero González, J. Puls, F. Najarro, and I. Brott. *A&A*, 537:A79, 2012. .
- N. Sanduleak. *Contributions from the Cerro Tololo Inter-American Observatory*, 89:1, 1970.
- M. F. Skrutskie, R. M. Cutri, R. Stiening, et al. *AJ*, 131:1163–1183, 2006. .
- S. A. Smee, J. E. Gunn, A. Uomoto, N. Roe, D. Schlegel, C. M. Rockosi, M. A. Carr, F. Leger, K. S. Dawson, M. D. Olmstead, J. Brinkmann, R. Owen, R. H. Barkhouser, K. Honscheid, P. Harding, D. Long, C. Loomis, R. H. Lupton, A. D. Montero-Dorta, and D. G. York. *AJ*, 146:32, 2013. .
- E. van Aarle, H. Van Winckel, T. Lloyd Evans, T. Ueta, P. R. Wood, and A. G. Ginsburg. *A&A*, 530:A90, 2011. .
- N. R. Walborn, I. D. Howarth, D. J. Lennon, P. Massey, M. S. Oey, A. F. J. Moffat, G. Skalkowski, N. I. Morrell, L. Drissen, and J. Wm. Parker. *AJ*, 123:2754–2771, 2002. .
- B. A. Whitney, M. Sewilo, R. Indebetouw, T. P. Robitaille, M. R. Meade, B. L. Babler, et al. *AJ*, 136:18–43, 2008. .
- D. Zaritsky, J. Harris, I. B. Thompson, and E. K. Grebel. *AJ*, 128:1606–1614, 2004. .

Appendix A: SDSS-V/BOSS observations and visit-resolved analysis

For [WSI2008] 778, four visit spectra (CATALOGID=63050395787470304) comprise six individual exposures with a total nominal exposure time of 5400 s. Three visits are available for MGSD LH 117-43A/140 (CATALOGID=63050395782984402), with a total nominal exposure time of 2700 s. For Sk -69 135 (CATALOGID=63050395785568943; SDSS_ID=91419225; Gaia Data Release 3 (DR3) source_id=4658095755697542528), the visits were obtained at modified Julian dates (MJDs) 60212, 60213, and 60235 and comprised four nominal 900-s exposures.

For the display combinations, each visit first received a broad 401-pixel multiplicative response correction before inverse-variance combination on the native logarithmic wavelength grid. Candidate artefacts were rejected only when the other visits agreed and the discrepancy was confined to an isolated run of one or two pixels; longer discrepant runs were retained and reported rather than clipped. Recurrent stellar, wind, and nebular features were therefore preserved. The BOSS wavelength scale is expressed in vacuum, and vacuum rest wavelengths were used throughout the radial-velocity analysis. Quantitative EW and velocity measurements were made from the individual visits, not from the display combination.

For the common stellar velocity, the available photospheric He II $\lambda\lambda 4200, 4542, 5412, 6683, 6891, 7178$ absorption lines were used where measurable. He II $\lambda 4686$ was excluded because of its wind sensitivity. Nitrogen-line velocities were used only as consistency checks, and the blended N III $\lambda\lambda 4634-4642$ complex was excluded because its centroid is unstable at BOSS resolution.

For Sk -69 135, He II $\lambda\lambda 4200, \lambda 4542$, and $\lambda 5412$ were fitted simultaneously in each visit with a common velocity shift and independent local continua, depths, and widths. Table A.2 gives the resulting visit velocities.

Table A.1. Stellar and nebular velocities.

Source	$v_{\text{He II}}$ (km s ⁻¹)	v_{neb} (km s ⁻¹)	Δv (km s ⁻¹)
[WSI2008] 778	295 ± 17	280.8 ± 4.3	+14 ± 18
MGSD LH 117-43A/140	240 ± 7	251.3 ± 4.4	-11 ± 8
Sk -69 135	277 ± 7	281 ± 19	-5 ± 20

Table A.2. Visit-resolved radial velocities of Sk -69 135.

MJD	$v_{\text{He II}}$ (km s ⁻¹)
60212	299.7 ± 13.8
60213	288.3 ± 15.3
60235	260.5 ± 11.2

Appendix B: Preparation of the LMC O2 comparison spectra

BI 237 and Brey 77 were observed with the ultraviolet-blue (UVB) arm of the Very Large Telescope (VLT)/X-shooter under European Southern Observatory (ESO) programme 106.211Z.001. Both products were obtained with the 0.8 × 11 arcsec slit and have a reference resolving power of $R = 6655$. The BI 237 and Brey 77 archive products are ADP.2021-03-08T12:59:03.267.fits and ADP.2021-01-11T12:13:26.255.fits, respectively.

The 30 Dor 016 template was constructed from stacked VLT/Fibre Large Array Multi Element Spectrograph (FLAMES)–GIRAFFE LR02 and LR03 products obtained under programme 182.D-0222(A). The LR02 setting covers 3959.99–4570.99 Å at $R = 6300$, whereas LR03 covers 4498.98–5077.98 Å at $R = 7500$. The two settings overlap between approximately 4499 and 4571 Å, allowing their relative continuum placement to be verified before the two spectral segments were joined.

Each template was shifted to its stellar rest frame and pseudo-continuum normalised over the blue-optical interval. Invalid pixels and isolated reduction artefacts were masked, while recurrent stellar, wind, and nebular features were retained. Because the archival templates have substantially higher spectral resolution than the BOSS data, they were convolved to the common wavelength-dependent instrumental width adopted for Fig. 1. The required Gaussian full width at half maximum (FWHM) of the kernel was calculated at each wavelength from

$$\text{FWHM}_{\text{ker}}(\lambda) = \left[\text{FWHM}_{\text{common}}^2(\lambda) - \text{FWHM}_{\text{temp}}^2(\lambda) \right]^{1/2},$$

where $\text{FWHM}_{\text{common}}(\lambda)$ is defined by the largest effective BOSS WRESL among the three science spectra. The corresponding common resolving power increases from approximately $R = 990$ at 4000 Å to $R = 1322$ at 5060 Å. No rotational broadening or adjustment of individual line strengths or profiles was applied. This homogenisation was used only for the morphological display in Fig. 1.

Appendix C: Statistical interpretation of the nitrogen diagnostic

Table C.1. Inverse-variance-weighted mean equivalent widths of the principal classification diagnostics measured in the individual SDSS-V/BOSS visits.

Feature	[WSI2008] 778 EW (Å)	MGSD LH 117-43A/140 EW (Å)	Sk -69 135 EW (Å)
N IV λ 4058	-0.75 ± 0.18	-0.82 ± 0.15	-0.38 ± 0.20
N III λ 4640	-0.17 ± 0.11	-0.27 ± 0.14	-0.17 ± 0.15
He I λ 4471	$+0.31 \pm 0.10$	$+0.23 \pm 0.12$	$+0.22 \pm 0.09$
He II λ 4542	$+0.67 \pm 0.15$	$+0.79 \pm 0.11$	$+0.67 \pm 0.13$
N V λ 4604	$+0.20 \pm 0.07$	$+0.31 \pm 0.08$	$+0.04 \pm 0.08$
N V λ 4620	$+0.27 \pm 0.08$	$+0.24 \pm 0.08$	-0.01 ± 0.13
He II λ 4686	-1.61 ± 0.37	$+0.80 \pm 0.11$	$+0.39 \pm 0.08$
H β	$+0.61 \pm 0.23$	$+1.90 \pm 0.17$	$+2.32 \pm 0.72$

Notes. Positive EWs denote absorption, whereas negative values denote emission. Each visit was measured independently after rest-frame correction. For [WSI2008] 778 and MGSD LH 117-43A/140, H β is affected by narrow nebular-emission infilling, as indicated independently by [O III] λ 4959, 5007; the quoted H β EWs should therefore not be interpreted as purely stellar line strengths.

For [WSI2008] 778 and MGSD LH 117-43A/140, both the measured N IV/N III ratios and the detection of N V λ 4604, 4620 support an O2 classification. Their weighted logarithmic ratios are 0.64 ± 0.29 and 0.48 ± 0.23 , respectively. For Sk -69 135, both the numerator and denominator are weak and the weighted ratio, 0.34 ± 0.44 , is correspondingly imprecise; neither N V component is significantly detected.

For Sk -69 135, the classification therefore rests mainly on the very-early-O morphology: N IV emission, weak N III, strong He II absorption, and the close similarity to BI 237, rather than on the exact central value of the formal N IV/N III ratio. The lack of a significant N V detection should not, however, be treated as proof of intrinsic absence, because shallow or rotationally broadened N V absorption can be difficult to recover at modest S/N (Walborn et al. 2002). Given these limitations, we adopt the conservative O2–3 subtype range for Sk -69 135 instead of an unqualified O2 classification.

Appendix D: Broad-band SED modelling and Gaia colour–magnitude diagram

D.1. OSTAR2002 SED fits and infrared diagnostics

The broad-band analysis uses the OSTAR2002 non-local thermodynamic equilibrium (NLTE), line-blanketed atmosphere grid (Lanz and Hubeny 2003). We adopt the half-solar grid at $\log g = 4.0$ and $d = 49.6$ kpc. Because the available optical and near-infrared photometry samples the Rayleigh–Jeans side of such hot SEDs, it is not used to determine T_{eff} . Instead, the five exact OSTAR2002 nodes at 45.0, 47.5, 50.0, 52.5, and 55.0 kK are sampled with equal probability. At each Monte-Carlo draw, catalogue uncertainties and an additional 2% flux floor are propagated while A_V and R_* are fitted jointly; $\log L/L_\odot$ and the Gaia broad-band extinctions are then derived for that draw.

We use the audited no- U photometric variants. For [WSI2008] 778 and Sk -69 135 the nominal optical constraints are the Massey V , $B - V$, and $V - R$ measurements (Massey 2002), together with Two Micron All Sky Survey (2MASS) JHK_s (Skrutskie et al. 2006); for MGSD LH 117-43A/140 we use Magellanic Clouds Photometric Survey (MCPS) BVI photometry (Zaritsky et al. 2004) and 2MASS. The same selection

procedure is applied to the early-O comparison stars: BI 237 uses MCPS BV and 2MASS JHK_s (the MCPS I datum is excluded after the leave-one-out consistency test identified it as the dominant discrepant measurement), whereas 30 Dor 016 uses Massey V , $B - V$, and $V - R$ together with 2MASS JHK_s . For [WSI2008] 778, an MCPS solution is propagated separately as a catalogue-systematic cross-check rather than merged into the nominal posterior. The U band is excluded throughout. Extinction within the modelled optical–near-infrared domain follows the LMC-average curve of Gordon et al. (2003). The Gaia G , G_{BP} , and G_{RP} extinctions are obtained by integrating the extinguished OSTAR2002 spectra through the corresponding passbands.

The nominal optical/near-infrared observables used in these fits are listed in Table D.1.

Figure D.1 compares the photospheric predictions with selected mid-infrared measurements. Spitzer Surveying the Agents of Galaxy Evolution (SAGE) photometry from the Infrared Array Camera (IRAC) is taken from the LMC survey (Meixner et al. 2006), AKARI points from the LMC point-source catalogue (VizieR J/AJ/144/179/lmccat01; Kato et al. 2012), and the 24- μm point of [WSI2008] 778 from the SAGE massive-YSO compilation (Whitney et al. 2008). The adopted Gordon et al. extinction curve is used only over its formal wavelength domain, ending at $\lambda = 3.33 \mu\text{m}$. Beyond this boundary we do not extrapolate an extinction law: the dashed curve is the maximum stellar photosphere obtained by setting $A_\lambda = 0$. Thus an observed point above that curve represents a conservative excess relative even to the largest allowed stellar continuum. No dust component is fitted.

The MIR measurements shown in Fig. D.1 are listed in Table D.2.

The posterior median ($A_V, R_*/R_\odot$) values are (0.74, 9.14), (0.60, 11.99), and (0.27, 12.94) for [WSI2008] 778, MGSD LH 117-43A/140, and Sk -69 135, respectively; these radii are conditional on the adopted 45–55 kK atmosphere range and are scaling checks rather than temperature determinations.

The qualitative result is not sensitive to the remaining temperature–radius covariance. For [WSI2008] 778, the excess grows strongly from the IRAC bands towards the AKARI and MIPS measurements, which can explain its previous infrared-selected classifications without implying that the optical spectrum itself is that of a YSO or post-AGB star. The excess may arise from circumstellar or local environmental dust, unresolved multiplicity, blending, or a combination of these effects. MGSD LH 117-43A/140 remains essentially photospheric over the available MIR coverage. For Sk -69 135, independent IRAC and AKARI measurements indicate only a mild systematic departure at 3–8 μm ; the previously apparent very large long-wavelength Wide-field Infrared Survey Explorer (WISE) excess is not retained because $W3$ and $W4$ are upper limits affected by persistence flags.

D.2. Gaia colour–magnitude diagram

The same Monte-Carlo realisations provide A_G , A_{BP} , and A_{RP} , which are combined with Gaia DR3 photometry (Gaia Collaboration 2023) to form

$$(G_{BP} - G_{RP})_0 = (G_{BP} - G_{RP}) - (A_{BP} - A_{RP})$$

and

$$M_{G,0} = G - A_G - 5 \log(d/10 \text{ pc}).$$

Table D.1. Optical and near-infrared photometry adopted in the nominal OSTAR2002 fits.

Source	Optical catalogue and native observables	2MASS JHK_s	Fit status
[WSI2008] 778	Massey: $V = 14.200 \pm 0.010$, $B - V = -0.040 \pm 0.010$, $V - R = +0.080 \pm 0.030$	$J = 14.314 \pm 0.036$, $H = 14.442 \pm 0.063$, $K_s = 14.476 \pm 0.094$	nominal
MGSD LH 117-43A/140	MCPS: $B = 13.443 \pm 0.029$, $V = 13.404 \pm 0.041$, $I = 13.570 \pm 0.062$	$J = 13.717 \pm 0.062$, $H = 13.757 \pm 0.051$, $K_s = 13.895 \pm 0.074$	nominal
Sk -69 135	Massey: $V = 12.980 \pm 0.010$, $B - V = -0.200 \pm 0.010$, $V - R = -0.070 \pm 0.020$	$J = 13.456 \pm 0.030$, $H = 13.566 \pm 0.041$, $K_s = 13.617 \pm 0.062$	nominal
BI 237	MCPS: $B = 13.790 \pm 0.023$, $V = 13.830 \pm 0.043$	$J = 13.950 \pm 0.027$, $H = 14.011 \pm 0.043$, $K_s = 14.039 \pm 0.072$	nominal
30 Dor 016	Massey: $V = 13.490 \pm 0.010$, $B - V = +0.040 \pm 0.010$, $V - R = +0.060 \pm 0.030$	$J = 13.384 \pm 0.041$, $H = 13.345 \pm 0.061$, $K_s = 13.356 \pm 0.057$	nominal

Notes. Magnitudes and colours are listed in the native catalogue observables actually fitted; in particular, Massey B and R magnitudes are not reconstructed or treated as independent data. The U band is excluded for all objects. For BI 237, the MCPS $I = 13.948 \pm 0.034$ measurement was examined but excluded from the nominal solution after the leave-one-out test identified it as the dominant discrepant datum. For [WSI2008] 778, the alternative MCPS measurements $B = 14.237 \pm 0.024$, $V = 14.363 \pm 0.029$, and $I = 14.181 \pm 0.026$ (with the same 2MASS counterpart) are propagated only as a catalogue-systematic cross-check and are not merged into the nominal posterior. A 2% flux floor is added in quadrature during fitting.

Table D.2. Selected mid-infrared diagnostic photometry displayed in Fig. D.1.

Source	SAGE/IRAC fluxes (mJy)	AKARI-LMC fluxes (mJy)	Other
[WSI2008] 778	3.6: 0.590 ± 0.022 ; 4.5: 0.480 ± 0.017 ; 5.8: 0.346 ± 0.020 ; 8.0: 0.330 ± 0.024	N3: 0.725 ± 0.055 ; S7: 0.271 ± 0.035 ; S11: 1.110 ± 0.080 ; L15: 1.700 ± 0.180 ; L24: 5.740 ± 0.330	MIPS 24: 4.690 ± 0.210
MGSD LH 117-43A/140	3.6: 0.802 ± 0.048 ; 4.5: 0.540 ± 0.028 ; 5.8: 0.334 ± 0.017 ; 8.0: 0.184 ± 0.018	–	–
Sk -69 135	3.6: 1.010 ± 0.030 ; 4.5: 0.668 ± 0.029 ; 5.8: 0.425 ± 0.018 ; 8.0: 0.261 ± 0.017	N3: 1.510 ± 0.120 ; S7: 0.298 ± 0.032	–

Notes. These measurements are diagnostic only and are excluded from the optical–near-infrared photospheric fit. SAGE/IRAC values are taken from the selected II/305/catalog entries, AKARI-LMC values from J/AJ/144/179/lmccat01, and the [WSI2008] 778 MIPS point from J/AJ/136/18/table12. When more than one entry from the selected catalogue product exists for a band, the plotted flux is the median and its adopted uncertainty is the larger of the median quoted uncertainty and the row-to-row scatter. AllWISE W3/W4 measurements of Sk -69 135 are not listed because they are upper-limit/persistence-affected entries and were not used as detections.

For the five O-star objects, the adopted CMD coordinates are the posterior medians of these quantities, and the 16th–84th percentile intervals are shown as asymmetric error bars in Fig. 2. BI 237 and 30 Dor 016 are processed through the same OSTAR2002/extinction framework and provide empirical very-early-O comparisons. The observed Gaia photometry and the final intrinsic CMD coordinates are tabulated in Table D.3.

Figure 2 compares the resulting positions with PARSEC v1.2S $Z = 0.006$ isochrones at 0.05, 0.1, 0.2, 0.5, 1, and 2 Myr (Bressan et al. 2012). At very-early-O temperatures, $G_{BP} - G_{RP}$ is only weakly sensitive to T_{eff} , so small horizontal reversals among adjacent young isochrones are not interpreted as a physical temperature inversion. Initial-mass labels on the 1 and 2 Myr curves are shown only as guides. The reddening vector corresponds to $A_V = 0.1$ mag and is derived with the same OSTAR2002 plus LMC-average extinction framework used for the homogeneous CMD corrections. The CMD is used to establish the luminosity regime rather than to infer precise ages or masses.

For comparison with binary-stripped objects, we considered the observational sample of intermediate-mass stripped helium stars analysed with CMFGEN by Götzberg et al. (2023). Stars 1–8, for which the stripped star dominates the optical light, span approximately $T_{\text{eff}} = 57\text{--}90$ kK and $\log L/L_\odot = 3.5\text{--}5.1$, so stripped stars can be as hot as, or hotter than, the earliest O stars. Applying the same hot-star synthetic Gaia-colour mapping used as an external diagnostic places this hot subsample at approximately $-0.56 \lesssim (G_{BP} - G_{RP})_0 \lesssim -0.51$; thus colour alone does not distinguish a hot stripped star from an early-O star. Their optical luminosities are nevertheless much lower. Using the pub-

lished AB photometry, extinctions, and Magellanic-Cloud distances gives approximately $-1.7 \lesssim M_{V,AB} \lesssim +0.8$ for Stars 1–8. Star 1, the optically brightest member of this hot stripped-star sample, is therefore shown in Fig. 2 as a single representative point. Its plotted Gaia coordinate is an approximate hot-star transformation of the published solution and is not part of the OSTAR2002 Monte-Carlo analysis.

The main distinction is therefore optical luminosity rather than intrinsic colour. The three science targets, with $M_{G,0} = -4.97$, -5.56 , and -5.79 mag, lie several magnitudes above the observed intermediate-mass stripped-star regime and occupy the same luminous blue domain as BI 237 and 30 Dor 016. Their CMD positions therefore strongly disfavour an interpretation in which the optically dominant components are ordinary intermediate-mass stripped helium stars, consistent with the small radii expected for binary-stripped products (Götzberg et al. 2018; Götzberg et al. 2023). This comparison does not exclude binary evolution: unresolved companions, past interaction, and more massive binary-evolution products remain possible. We therefore use the CMD to establish the luminosity regime, not to assign precise ages, initial masses, or single-star evolutionary histories.

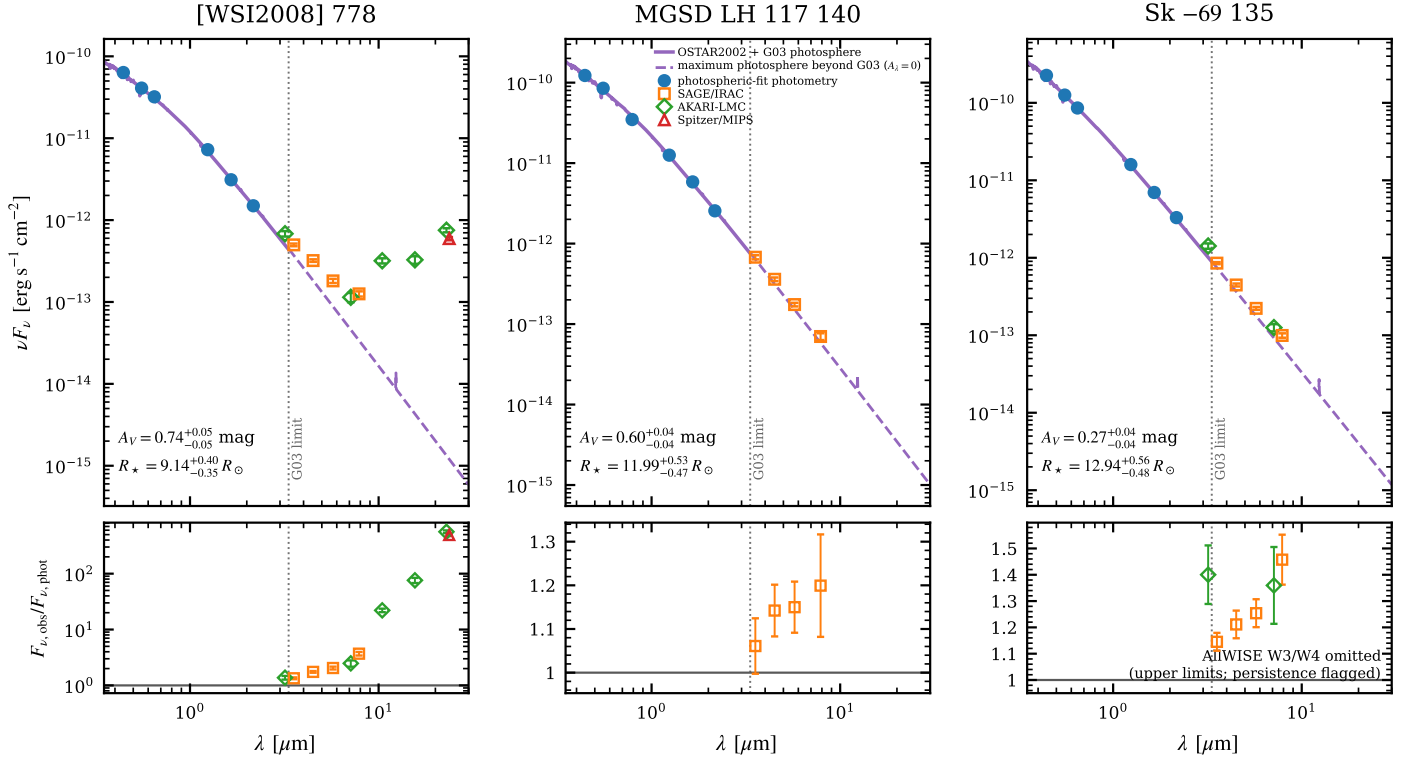


Fig. D.1. Broad-band SEDs of [WSI2008] 778 (left), MGSD LH 117-43A/140 (middle), and Sk -69 135 (right). Upper panels show νF_ν , and the lower panels show the observed-to-photospheric flux ratio. Within the formal Gordon et al. (2003) domain, the solid curve is the median OSTAR2002 photosphere reddened with their LMC-average curve, and the shaded interval represents the 16th–84th percentiles of the Monte-Carlo model. The vertical dotted line marks $3.33\,\mu\text{m}$. At longer wavelengths, the dashed curve is the conservative maximum photosphere obtained with $A_\lambda = 0$; no MIR extinction law is extrapolated. Filled circles are the optical/near-infrared constraints used in the photospheric fit, while open symbols show the selected SAGE/IRAC, AKARI-LMC, and, where available, Spitzer/MIPS (Multiband Imaging Photometer for Spitzer) measurements. [WSI2008] 778 displays a strong rising MIR excess; MGSD LH 117-43A/140 remains close to the photosphere; Sk -69 135 shows a modest 3–8 μm excess. AllWISE W3 and W4 for Sk -69 135 are not used because the catalogue reports upper-limit photometric-quality flags and persistence contamination in both bands.

Table D.3. Gaia DR3 photometry and extinction-corrected CMD coordinates.

Source	Gaia DR3 source_id	G	G_{BP}	G_{RP}	$(G_{BP} - G_{RP})_0$	$M_{G,0}$
[WSI2008] 778	4658432721684644480	14.256787	14.193605	14.279823	$-0.509^{+0.033}_{-0.034}$	$-4.969^{+0.052}_{-0.050}$
MGSD LH 117-43A/140	4657383027359753728	13.519092	13.427448	13.600076	$-0.514^{+0.025}_{-0.025}$	$-5.563^{+0.050}_{-0.050}$
Sk -69 135	4658095755697542528	12.970034	12.854306	13.174096	$-0.478^{+0.025}_{-0.025}$	$-5.788^{+0.059}_{-0.059}$
BI 237	4660109236409051392	13.825301	13.778407	13.879809	$-0.524^{+0.023}_{-0.021}$	$-5.400^{+0.047}_{-0.043}$
30 Dor 016	4657690620070706432	13.479229	13.482241	13.430018	$-0.562^{+0.024}_{-0.024}$	$-6.081^{+0.049}_{-0.049}$
Göteborg Star 1	—	—	—	—	≈ -0.560	≈ -1.80

Notes. The Gaia magnitudes are the observed DR3 mean photometry. For the five O-star objects, the intrinsic colour and absolute magnitude are posterior medians; superscripts and subscripts give the 84th-minus-median and median-minus-16th-percentile intervals, respectively, from the homogeneous OSTAR2002/extinction Monte-Carlo propagation. The Göteborg Star 1 values were obtained using the approximate Gaia-equivalent external transformation described in the text and do not carry a homogeneous Monte-Carlo interval.

Appendix E: Acknowledgements

This research is based in part on observations collected at the European Southern Observatory under ESO programmes 106.211Z.001 and 182.D-0222(A), and on data obtained from the ESO Science Archive Facility. The archival spectra used in this work comprise the VLT/X-shooter observations of BI 237 and Brey 77 obtained under programme 106.211Z.001 and the VLT/FLAMES–GIRAFFE observations of 30 Dor 016 obtained under programme 182.D-0222(A).

Funding for the Sloan Digital Sky Survey V has been provided by the Alfred P. Sloan Foundation, the Heising-Simons Foundation, the National Science Foundation, and the Partici-

pating Institutions. SDSS acknowledges support and resources from the Center for High-Performance Computing at the University of Utah. SDSS telescopes are located at Apache Point Observatory, funded by the Astrophysical Research Consortium and operated by New Mexico State University, and at Las Campanas Observatory, operated by the Carnegie Institution for Science. CRZ acknowledges financial support from the UNAM-DGAPA-PAPIIT IN107226. J.G.F.-T. gratefully acknowledges support from ANID Fondecyt Regular No. 1260371. The SDSS web site is www.sdss.org.

SDSS is managed by the Astrophysical Research Consortium for the Participating Institutions of the SDSS Collaboration, including the Carnegie Institution for Science, Chilean National

Time Allocation Committee (CNTAC) ratified researchers, Caltech, the Gotham Participation Group, Harvard University, Heidelberg University, The Flatiron Institute, The Johns Hopkins University, L'Ecole polytechnique fédérale de Lausanne (EPFL), Leibniz-Institut für Astrophysik Potsdam (AIP), Max-Planck-Institut für Astronomie (MPIA Heidelberg), Max-Planck-Institut für Extraterrestrische Physik (MPE), Nanjing University, National Astronomical Observatories of China (NAOC), New Mexico State University, The Ohio State University, Pennsylvania State University, Smithsonian Astrophysical Observatory, Space Telescope Science Institute (STScI), the Stellar Astrophysics Participation Group, Universidad Nacional Autónoma de México, University of Arizona, University of Colorado Boulder, University of Illinois at Urbana-Champaign, University of Toronto, University of Utah, University of Virginia, Yale University, and Yunnan University.